

Hybrid Agentic Training Framework for Neural Networks on MedMNIST

Integrated Project Proposal and Pre-Project Documentation

Acronym: **HAT-MedMNIST**

Project acronym	HAT-MedMNIST
Planned duration	12 months
Target output	Validated software MVP (research / engineering demonstrator)
Target TRL	TRL 7 — system prototype demonstrated in a representative operational environment
Development model	Hybrid Agile / stage-gate
Core design principle	Maximum cohesion, minimum coupling
Distinctive approach	Hybrid agents (deterministic execution + open-model reasoning); no vendor lock-in
Domain	Agentic AI / medical-image benchmarks (non-clinical)

PART I — Project Team and Responsibilities

The team is organised around three reference roles (names to be inserted before submission).

Role	Team Member	Main Responsibilities
Technical Responsible (TR)	Todisco Giuseppe	Owns technical architecture, scientific soundness, agent design, representation and training strategy, integration, and technical acceptance of deliverables.
Project Manager (PM)	Verrilli Stefano	Coordinates scope, schedule, resources, costs, reporting, cross-activity dependencies, risk management, and stakeholder communication.
Reviewer (REV)	Dolatabadi Mohammad	Independently reviews work packages, activity lines, deliverables, KPIs, traceability, reproducibility, and coherence with the state of the art and TRL. Also embodied in the project as the automated consistency / anomaly agent (WP6).

PART II — General Project Proposal

1. Project Data and Abstract

The project develops and validates a **modular, hybrid agentic framework** for training neural networks on the MedMNIST suite. Distinct software agents own bounded prerogatives across the training lifecycle — data ingestion, profiling and **ambiguity handling**, **prior-art scouting**, **representation design**, experiment design, training, evaluation, **abstention on uncertain cases**, enrichment, and reporting — while an independent **consistency / anomaly agent** audits every stage and can veto promotion. The architecture follows maximum cohesion and minimum coupling: each agent has one responsibility, and agents interact only through explicit, versioned artefacts.

The approach is **hybrid**: execution steps that must be reproducible (data, splits, training, metrics) are deterministic, while genuine judgement points (ambiguity assessment, representation design, configuration choice, consistency review) use reasoning from an **open large language model** served

through a provider-agnostic interface, with a deterministic heuristic fallback. This deliberately avoids any model “owned by a company” and keeps the whole pipeline runnable offline and reproducibly.

The result is a software MVP that orchestrates repeatable training experiments, compares agent-driven decisions with a conventional baseline, handles the “no confident finding” case, and generates auditable evidence. The project is a research and engineering demonstrator on benchmark data; clinical deployment, diagnosis, and patient-facing use are out of scope.

2. Positioning with Respect to the State of the Art

Conventional training pipelines are typically tightly coupled scripts or notebooks in which data preparation, model configuration, optimisation, validation, and reporting are interdependent, limiting maintainability, reproducibility, and auditability. Recent LLM-agent frameworks, conversely, often push non-determinism into every step, harming reproducibility and creating dependence on a single proprietary model.

HAT-MedMNIST advances beyond both by combining a **role-based agentic orchestration layer** with **explicit artefact contracts**, a **hybrid deterministic / open-model design** that confines non-determinism to declared decision points, **full decision provenance**, and an **independent automated reviewer** that verifies outputs. Progress is assessed not only by predictive performance but by engineering quality: modularity, reproducibility, robustness, auditability, and the ability to replace or upgrade one agent without redesigning the system. The Reviewer will specifically assess coherence of the claimed advancement with the state of the art.

3. Scientific and Technological Objectives

- Design a multi-agent architecture for neural-network training with explicit, cohesive, weakly-coupled responsibilities communicating only through typed artefacts.
- Introduce a **representation-architect agent** that reasons about the best input representation (normalization, label-preserving augmentation) for the task, records the decision, and applies it deterministically.
- Introduce a **prior-art scouting agent** that searches the literature / web for how the problem is addressed, and distils **cited, cached, reproducible** recommendations into the representation and experiment-design decisions.
- Introduce a **hybrid reasoning seam** using open models, with heuristic fallback, so decisions are reasoned yet reproducible and provider-independent.
- Implement a reproducible data and experiment pipeline for selected MedMNIST tasks, including **ambiguity handling** and leakage-safe splits.
- Handle the “**no confident finding**” case via an abstention / out-of-distribution agent.
- Provide an **independent consistency / anomaly agent** (automated reviewer) that audits every stage and can veto promotion.
- Compare agentic and conventional workflows under common conditions with transparent baselines, and deliver an integrated MVP demonstrated at TRL 7.

4. Proposal Themes

- **Scientific development:** study of role-specialised hybrid agents and their effect on training quality, repeatability, and decision traceability, including the representation-design and abstention mechanisms.

- **Competitive industrial development:** a differentiated orchestration capability that reduces experimentation time and improves maintainability and auditability, without proprietary-model lock-in.
- **Pre-competitive enabling development:** reusable artefact contracts, evaluation protocols, and governance components (the consistency / anomaly agent) applicable beyond MedMNIST.
- **Technology transfer and valorisation:** transition from MVP to configurable training platforms, validation services, and domain-specific extensions.

Synergies with other projects are not foreseen at this stage.

5. Expected Results and Main Deliverables

Objectives describe the intended capability; deliverables are the concrete, reviewable outputs that demonstrate it.

- **D1** — Requirements, modular architecture & interface specification, governance rules, feasibility study, baseline protocol (WP1).
- **D2** — Reproducible data-understanding & representation pipeline: profiling, ambiguity flags, prior-art brief (cited & cached), representation plan, leakage-safe splits (WP2).
- **D3** — Hybrid multi-agent orchestration engine with artefact contracts, decision provenance, and a data-, representation- & prior-art-driven model-design loop (WP3).
- **D4** — Evaluation, data enrichment, benchmark comparison, ablations, reproducibility package (WP4).
- **D5** — Integrated MVP with abstention / OOD handling, demonstration, documentation, final validation dossier, exploitation roadmap (WP5).
- **D6** — Consistency / anomaly governance agent + traceability & reproducibility evidence, cross-cutting (WP6).

6. Technical-Scientific Feasibility, Impact, and Valorisation

Feasibility is supported by a bounded benchmark suite, deliberately small models (the contribution is the orchestration, not the classifier), modular components with automated tests, and measurable acceptance criteria. Work follows a hybrid Agile / stage-gate model: short iterations inside each WP, with formal gate reviews controlling architecture, data readiness, model validation, MVP integration, and final acceptance.

Expected impact: less manual coordination and rework, stronger reproducibility, faster diagnosis of training failures via error analysis and the anomaly agent, clearer allocation of responsibility, and freedom from proprietary-model dependence. The industrial value is a reusable software asset rather than a one-off experiment. Valorisation follows an **MVP → higher-gain-product** cycle: the MVP proves the core orchestration, representation, abstention, and governance capabilities; productisation may add scalable compute, domain adapters, enterprise governance, integration APIs, monitoring dashboards, and support services. Exploitation models include licensed software, managed experimentation services, and reusable internal platforms.

7. Legal, Ethical, and Governance Framing

Only data and tools whose licences permit the planned research use will be used; MedMNIST provenance, licences, and generated artefacts are documented. Data minimisation, access control, secure storage, reproducibility, model limitations, bias analysis, and human review are addressed in the project records. Because the work is a research demonstrator on benchmark data, no clinical

claims are made and no diagnostic decision is delegated to the system; the abstention agent explicitly routes uncertain cases to human review.

8. Verification of Project Outcomes and TRL

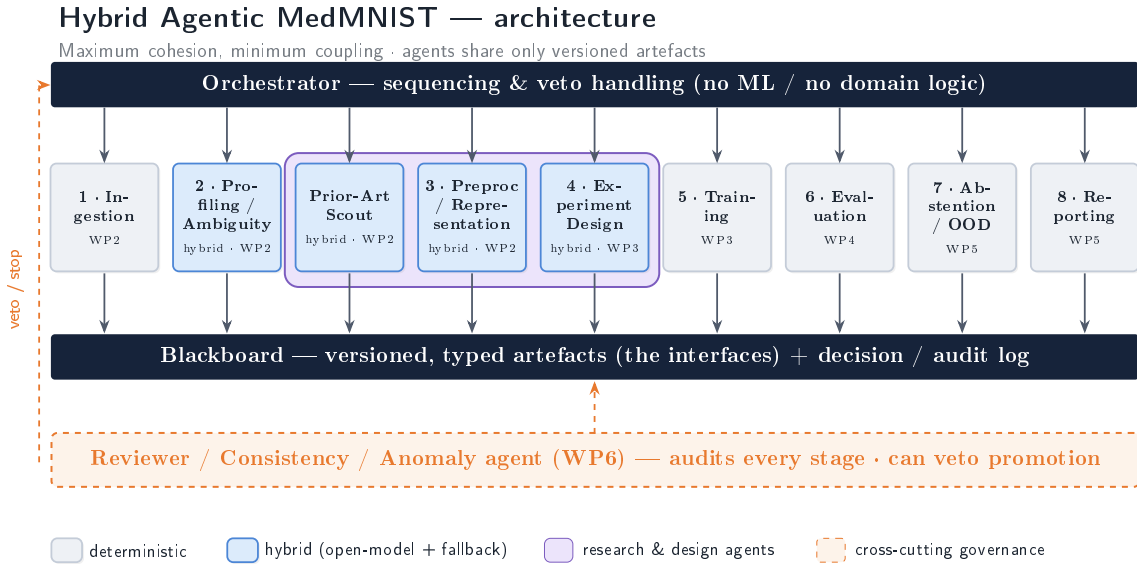
Outcome verification is based on the WP structure: each activity line produces a deliverable and one or more KPIs. The Reviewer verifies completeness, objective evidence, test results, traceability to requirements, reproducibility, and consistency between results and declared TRL.

The target is **TRL 7**: a system prototype demonstrated in a representative operational environment. Here the representative environment is a realistic end-to-end software workflow — automated ingestion, prior-art scouting with cached provenance, agent-architected representation, coordinated agent decisions with logged provenance, controlled training, evaluation against a baseline, abstention on uncertain cases, cross-cutting consistency auditing with veto, recovery from expected faults, and an end-to-end demonstration executed by a user other than the implementer. This target does not imply clinical validation or certification.

PART III — Project Structure, Work Plan, and Costs

9. System Architecture

Agents never call one another; they read and write **versioned, typed artefacts** on a shared Blackboard, and a stateless Orchestrator sequences them. This makes “minimum coupling” concrete: any agent can be replaced or upgraded without touching another. The cross-cutting Reviewer agent (WP6) audits each artefact after every stage and can veto promotion to the next. The **prior-art scouting agent** runs after profiling and writes a cited, cached *research brief* that informs the representation (step 3) and experiment-design (step 4) decisions, without itself training or selecting.



Each blue (hybrid) agent uses open-model reasoning with a deterministic heuristic fallback and records the provenance of every decision. The prior-art scout informs the representation and experiment-design decisions; the Reviewer audits every stage and can veto promotion.

10. Work Packages and Activity Lines

Activity lines may interact. Dependencies and exchanged artefacts are made explicit so that data preparation, representation, training, evaluation, and reporting remain independently testable while supporting the end-to-end process.

Table 1. Work-package overview

WP	Title	Purpose	Deliv.	WP KPI
WP1	Requirements & Architecture	Scope, roles, artefact contracts, governance, feasibility, baseline & acceptance criteria (Agile).	D1	100% critical requirements traced; architecture & coupling review approved.
WP2	Data Understanding, Representation & Prior-Art	Ingestion, profiling, ambiguity handling, prior-art scouting, representation design, leakage-safe splits.	D2	$\geq 99\%$ samples profiled; splits reproduced from seed; representation plan validated; 100% scouted recommendations cited & cached.
WP3	Hybrid Agentic Orchestration & Model Design	Orchestrator, contracts, provenance, data-, representation- & prior-art-driven config and training.	D3	All planned agents integrated; $\geq 95\%$ nominal runs without manual repair; 0 direct inter-agent calls.
WP4	Evaluation, Enrichment & Comparison	Metrics, data enrichment, baseline comparison, ablations, error analysis.	D4	Baseline met or exceeded on primary metric; repeated-run variance within threshold.
WP5	MVP Integration & Validation	Integrate, add abstention / OOD, demonstrate, document, assess TRL, costs, exploitation.	D5	End-to-end acceptance pass; independent demonstration completed; TRL 7 evidence approved.
WP6	Consistency & Anomaly Governance	Automated reviewer auditing all stages with veto; traceability & reproducibility (cross-cutting).	D6	$\geq 90\%$ injected inconsistencies detected; 100% agent decisions logged; full run reproducible.

Table 2. Activity lines, deliverables, links, and KPIs

LA	Responsibility	Activity / Deliverable Title	Type	Linked LA	Key Performance Indicator	Verification Evidence
1.1	PM	Requirements, scope & acceptance plan (functional: accuracy, precision, decision boundaries; non-functional: interpretability, agent-duty disentanglement, usability by a non-technical team)	Document	All	$\geq 90\%$ stakeholder requirements agreed; 100% critical requirements owned	Approved requirement matrix & review minutes
1.2	TR	Modular agent architecture & artefact-contract specification	Design	2.x, 3.x	All agent duties & interfaces documented; coupling review passed (0 direct inter-agent calls)	Architecture document & contract tests
1.3	REV / PM	Feasibility study vs state of the art	Report	1.1, 1.2	Feasibility approved; functional & non-functional baseline defined	Feasibility dossier & review sign-off
2.1	TR	Ingestion + profiling / ambiguity agent (DataProfile)	Software	1.2, 3.2	$\geq 99\%$ samples profiled; ambiguity flags produced; deterministic ingestion	Pipeline logs & data-lineage report
2.2	TR	Representation-architect preprocessing (RepresentationPlan) + leakage-safe splits	Software	2.1, 3.2	Train-only normalization stats; augmentations from label-preserving set; splits reproduced from seed	Representation plan artefact & rerun evidence
2.3	TR	Prior-art scouting agent (ResearchBrief) with source cache & provenance	Software	2.1, 2.2, 3.2, 6.1	100% recommendations cited; sources snapshotted & checksummed; brief reproducible from cache; 0 unresolved citations	ResearchBrief artefact & source cache
2.4	REV	Data quality, leakage, representation & prior-art audit	Report	2.1, 2.2, 2.3, 6.1	No unresolved leakage; representation plan independently validated; scouted citations verified to resolve	Audit checklist & rerun evidence
3.1	TR	Orchestrator + Blackboard + artefact contracts (core engine)	Software	1.2, 2.x	100% planned agents integrated; $\geq 95\%$ nominal runs without manual repair	Integration-test report & trace logs
3.2	TR	Experiment-design (config from profile + representation) + training	Software	2.1, 2.2, 4.1	Config provenance logged; training reproducible from seed + config	Experiment registry & trained artefacts
3.3	REV	Intermediate review gate + fault handling / fallback	Software / Test	3.1, 6.1	Veto mechanism functional; $\geq 90\%$ recovery from modelled faults; 100% decisions logged	Fault-injection report & audit logs
4.1	TR	Evaluation agent + conventional baseline	Model / Report	3.2	Baseline met or exceeded on agreed primary metric; common seeds used	Evaluation report & baseline comparison
4.2	TR	Data enrichment (multi-source / augmentation) with provenance	Software / Dataset	2.2, 4.1	Enrichment provenance tracked; ≥ 1 enrichment scenario compared	Enrichment lineage & comparison

Table 2 (continued)

LA	Responsibility	Activity / Deliverable Title	Type	Linked LA	Key Performance Indicator	Verification Evidence
4.3	REV / TR	Robustness, ablation (incl. representation on/off) & error analysis	Report	4.1, 4.2	≥ 3 ablation scenarios; repeated-run variance within threshold	Ablation report & reproducibility package
5.1	TR	Integrated MVP + abstention / OOD (“no finding”) handling	Prototype	3.3, 4.3	End-to-end run by an independent user; abstention lowers error on the covered set	Acceptance test & demonstration record
5.2	PM / REV	Final documentation, TRL assessment, cost & exploitation roadmap	Document	All	100% mandatory deliverables accepted; no open critical risk; TRL 7 evidence mapped	Final dossier & signed review checklist
6.1	REV	Consistency / anomaly agent monitoring all stages, with veto	Software	All	$\geq 90\%$ injected inconsistencies detected; 100% agent decisions audited	Anomaly reports & injection evidence
6.2	PM	Traceability & reproducibility package	Document	All	Full run reproducible from stored artefacts, config & seeds	Reproducibility rerun & artefact registry

11. Indicative Gantt Plan

The 12-month plan is indicative and can be adjusted during contract negotiation. Overlap is intentional where activity lines exchange validated intermediate artefacts; WP6 is cross-cutting.

WP / Month	1	2	3	4	5	6	7	8	9	10	11	12	
WP1	•	•											
WP2		•	•	•									
WP3			•	•	•	•	•						
WP4					•	•	•	•	•				
WP5								•	•	•	•	•	
WP6		•	•	•	•	•	•	•	•	•	•	•	

12. Risk Plan

Table 3. Principal project risks

Risk	Impact	Prob.	Effects and Rationale	Actions and Solutions	Linked LA
Data leakage / invalid split	High	Medium	Inflated performance, invalid conclusions.	Train-only normalization stats, seeded splits, leakage tests, reviewer approval before training.	2.2, 2.4, 6.1
Non-deterministic LLM reasoning	Medium	Medium	Irreproducible decisions, unstable results.	Temperature 0, deterministic heuristic fallback, provenance logging, guardrails on agent outputs.	3.2, 6.1
Unreliable / hallucinated prior-art sources	Medium	Medium	Ungrounded or non-reproducible design suggestions; dependence on live web.	Snapshot & checksum every source, citation-resolves check, temperature 0, heuristic fallback, reviewer veto on the brief.	2.3, 2.4, 6.1
Agent interaction / contract failure	High	Medium	Deadlock, inconsistent decisions, incomplete runs.	Typed artefact contracts, timeouts / retries, orchestrator veto, reviewer invariants, fault-injection tests.	3.1, 3.3
Representation choice harms task	Medium	Medium	Unsafe augmentation or normalization degrades learning.	Label-preserving guardrail set, independent reviewer re-validation, representation on/off ablation.	2.2, 2.4, 4.3
Performance below baseline	High	Medium	Weak scientific / industrial value.	Baseline-first strategy, ablations, alternative agent policies, documented negative results.	4.1, 4.3
Data loss or corruption	High	Low	Lost experiments, non-reproducibility, delay.	Versioned artefacts, checksums, immutable registry, restore tests.	2.1, 3.2
Proprietary-model / vendor lock-in	Medium	Low	Cost, dependence, non-portability.	Open-model via OpenAI-compatible seam; no proprietary dependency; heuristic fallback.	3.2, 6.1
Unmodelled critical event	High	Low	Potential loss of funding, evidence, or acceptance.	Contingency reserve, incident log, change-control, escalation path, periodic risk review.	All

13. Cost Structure and Budget Framework

Total budget to be defined; shares are a planning baseline, replaceable with contractual amounts. Compute is comparatively low because models are deliberately small and reasoning uses a local open model.

Cost Category	Indic. Share	Main Cost Drivers	Value / Industrial Rationale
Personnel	55%	Architecture, data & representation engineering, agent development, experiments, PM, review.	Reusable know-how, software assets, documentation.
Compute & storage	18%	Training runs, local open-model serving, experiment registry, backups, test environments.	Repeatability, comparison, representative demonstration.
Software & tools	5%	Development, monitoring, collaboration; open-source-first.	Reduced implementation time; no licence lock-in.
Validation & quality	10%	Independent review, test design, reproducibility, fault injection, anomaly-agent evaluation.	Higher acceptance probability; protects credibility.
Documentation & dissemination	5%	Technical documentation, demonstration material, final reporting.	Supports transfer, reuse, exploitation.
Contingency	7%	Unplanned compute, rework, integration or risk response.	Protects schedule and deliverable completion.

14. Final Acceptance Criteria

- All mandatory deliverables complete, versioned, and linked to requirements and activity lines.
- KPIs measured with documented procedures and objective evidence.
- The hybrid agentic workflow demonstrated end-to-end on selected MedMNIST tasks in a representative software environment, including abstention on uncertain cases.
- Results reproducible from stored configurations, code, data references, and seeds; every agent decision carries logged provenance.
- No critical risk remains without an approved mitigation or acceptance decision.
- The Reviewer confirms coherence of the scientific advancement, industrial potential, and TRL 7 evidence.

This document constitutes the initial pre-project documentation and provides the baseline for detailed planning, implementation, monitoring, and final review.